

PUBLIC RESEARCH / PORTFOLIO PROJECT

CYBERSECURITY ASSESSMENT REPORT

Real Estate Referral Platform — Anonymized Case Study

External Cybersecurity & Privacy Posture Assessment

Platform Assessed:	Real estate referral platform (name and URL withheld)
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Document Control

Version	Date	Author	Description
1.0	June 15, 2026	Kyle Dwyer	Initial cybersecurity and privacy assessment of the platform
1.1	July 17, 2026	Kyle Dwyer	Updated cybersecurity assessment of the platform
1.2	July 19, 2026	Kyle Dwyer	Sanitized for public release — removed identifying details and reproducible technical specifics
1.3	July 20, 2026	Kyle Dwyer	Anonymized for public release — platform name, URL, and other business-identifying details withheld

Classification: Public. This assessment is published as a portfolio/research project with the permission of the assessed organization. This version has been sanitized for public release: identifying details, proprietary information, and reproducible technical specifics have been removed or generalized.

Executive Summary

Kyle Dwyer conducted this assessment to evaluate the cybersecurity posture, privacy practices, and overall trustworthiness of an online real estate referral and matching platform (name and web address withheld for this published case study). The platform connects consumers with licensed real estate professionals, with a stated mission of increasing visibility for underrepresented real estate professionals.

The assessment focused on publicly accessible portions of the platform, examining website architecture, privacy practices, legal documentation, authentication controls, input validation mechanisms, and organizational trust indicators. Testing relied on passive reconnaissance and limited user-side interaction with publicly available forms and authentication systems; no attempts were made to bypass controls, gain unauthorized access, or exploit vulnerabilities.

0 Critical Vulnerabilities Identified	C- Security Header Score (45/100)	Moderate Overall Risk Rating
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Several positive security controls were identified, including HTTPS encryption with HSTS, Google reCAPTCHA, generic authentication error messaging, publicly accessible privacy and legal documentation, and basic form validation. No critical vulnerabilities were discovered during testing, and the organization demonstrated transparency through active social media channels and publicly available contact information.

The most significant opportunity for improvement lies in realtor identity verification and onboarding governance: accounts can currently be created and made publicly searchable without email verification, license validation, or administrative review. Multi-factor authentication, stronger password-reset protections, and a formal vulnerability disclosure process would further strengthen the platform's security posture.

Overall, the platform maintains an acceptable baseline level of security for a public-facing referral platform and demonstrates several industry-standard practices that help protect users and establish organizational credibility.

Company Background

Note on Anonymization

This version of the report has been anonymized for public portfolio use. The platform's name, web address, and other business-identifying details have been withheld or generalized. General industry context is retained where necessary to preserve the risk narrative and rationale behind the findings.

The assessed platform is an online service designed to connect homebuyers, sellers, and investors with qualified real estate professionals. According to its publicly published mission statement, the platform was created to increase visibility and accessibility for underrepresented real estate professionals while remaining open to consumers and agents of all backgrounds.

The homepage emphasizes the platform's mission of connecting users with trusted real estate professionals nationwide while supporting representation within the real estate industry. The platform clearly states

throughout the site that it functions as a marketing, referral, and matching service rather than a licensed real estate brokerage.

The Terms and Conditions identify the platform as owned and operated by a licensed real estate professional based in the Commonwealth of Virginia. Referral compensation related to real estate transactions is disclosed as being handled through the appropriate licensed brokerage in accordance with applicable law — a distinction that demonstrates legal transparency regarding the organization's role in real estate transactions.

Assessment Methodology

The assessment used publicly accessible information and non-intrusive testing techniques only. No attempts were made to bypass authentication controls, gain unauthorized access, exploit vulnerabilities, or perform denial-of-service testing. Testing activities were grouped into five areas:

Website Reconnaissance

- Review of public webpages and navigation structure
- Identification of publicly available functionality
- Examination of service offerings

Documentation Review

- Privacy Policy and Terms and Conditions analysis
- Review of legal disclosures and organizational transparency

Authentication Testing

- Login page review and credential validation testing
- User enumeration testing
- Authentication workflow analysis

Form Security Testing

- Contact form and input validation testing
- Basic cross-site scripting (XSS) testing
- Error message analysis

Trust Verification

- Social media verification
- Contact information and public business presence review

Website Architecture and Public Exposure

Several publicly accessible sections of the website were identified, including the homepage, buying and selling resources, an events section, a "Kitchen Table" educational section, an About Us page, a contact page, a login portal, and agent search functionality.

The site follows a relatively straightforward architecture designed primarily for consumer interaction and lead generation. Users can either browse agents independently or participate in a guided matching process that collects information about their real estate goals and preferences. The navigation structure is intuitive, reducing user confusion and supporting a positive user experience.

From a cybersecurity perspective, limiting unnecessary complexity within public-facing applications reduces attack surface and decreases the likelihood of misconfiguration.

Privacy and Data Handling Assessment

The platform provides a publicly accessible Privacy Policy that clearly outlines its data collection and processing practices.

Information Collected

- Contact information: name, email address, phone number, and additional voluntarily provided details
- User profile information: professional details, brokerage affiliations, and agent licensing information
- Usage information: IP addresses, browser and device information, log files, and cookie data
- Transaction information: billing and payment data, processed through third-party payment providers

Payment information is explicitly stated to be processed by third-party providers rather than stored directly by the platform, reducing its exposure to sensitive financial data and lowering the risk associated with storing payment card information.

Information Usage and Sharing

Collected information may be used to operate and improve the platform, facilitate connections between consumers and agents, communicate with users, personalize content, and monitor platform integrity and security — purposes consistent with the services offered.

Information may be shared with real estate professionals selected by users, service providers supporting platform operations, when required by law, and during business transfers such as mergers or acquisitions. This level of transparency contributes positively to user trust.

User Privacy Rights

Users are informed of rights to access, correct, and request deletion of their information, and to opt out of marketing communications — demonstrating awareness of modern privacy expectations and evolving data protection requirements.

Legal and Compliance Assessment

The platform provides a comprehensive Terms and Conditions page outlining expectations for users and legal limitations associated with the service.

- Eligibility: users must be at least 18 years old and legally capable of entering into binding agreements
- Platform purpose: the platform repeatedly clarifies that it is not a licensed brokerage and does not provide legal, financial, or transaction management services
- User responsibilities: provide accurate information, maintain account confidentiality, use the platform appropriately, and avoid abusive or fraudulent behavior
- Intellectual property: site content, graphics, logos, and trademarks remain the platform owner's property unless otherwise noted
- Governing law: disputes are governed by the laws of the Commonwealth of Virginia

The inclusion of detailed legal documentation contributes positively to organizational transparency and demonstrates a mature approach to risk management.

Authentication and Access Control Assessment

Authentication testing was performed against the platform's realtor login and account management system to evaluate how accounts are created, authenticated, recovered, and managed after onboarding, including activation workflows, password recovery, profile management, and realtor identity validation.

The platform provides a standard login interface with email/password authentication, password recovery, account management settings, and profile customization. After authentication, users can access profile settings, password management, resource pages, and billing/payment navigation.

Key Observation — Weak Onboarding Controls

Account registration and onboarding controls appeared significantly weaker than expected for a platform built around professional identity and consumer trust. A realtor account was successfully created using placeholder information and became active immediately, with no visible verification requirements or administrative approval process. Email verification was not enforced before activation — a verification email was sent, but its necessity was not communicated to the user, and login access was granted immediately after registration.

A test profile using placeholder identity details appeared within the realtor search functionality shortly after account creation, despite no evidence of identity verification, brokerage confirmation, or license validation. Because consumers may reasonably assume listed realtors have undergone some level of professional vetting, unverified accounts becoming publicly searchable introduces opportunities for impersonation, misinformation, fraudulent listings, and abuse of platform credibility.

Most profile information — including name, phone number, brokerage information, license-related information, service areas, and profile images — could be modified after account creation without secondary verification. Additional licenses could also be added without visible validation or review, allowing inaccurate or misleading professional claims to appear publicly.

Profile image upload functionality appeared operational, with no immediate upload failures observed; deeper backend testing of upload handling was outside the scope of this assessment.

Password recovery generates a numeric reset code delivered via email. While functional, rate limiting, account lockout protections, and monitoring controls were not visibly observable during limited testing.

Basic HTML and script injection attempts through user-input fields did not execute, suggesting some level of frontend sanitization or output encoding is already in place, though additional testing would be required to fully evaluate cross-site scripting protections across the entire application.

Overall, the authentication system is functional from a usability perspective but currently places substantial trust in self-submitted information without stronger verification or approval controls.

Identity and Account Security Assessment

Because the platform's core value proposition depends on connecting consumers with trustworthy professionals, identity verification should be treated as a core security and business requirement rather than a routine user-management feature.

Realtor accounts could be created with arbitrary or placeholder information and activated immediately without manual approval or verification, and no email verification enforcement was observed. From a business-risk perspective, this creates several concerns:

- Unverified users may publicly present themselves as legitimate professionals
- Fraudulent accounts may damage platform credibility
- Consumers may unknowingly interact with fake profiles
- Bad actors could abuse the platform for spam, impersonation, or social engineering
- Trust in realtor recommendations may decrease over time

Exposure extends beyond internal account management because unverified accounts appear directly in public search results. Testing did not identify visible controls for state license validation, brokerage confirmation, manual account approval, email ownership verification, or administrative onboarding review.

Users could also modify most profile information after account creation without additional verification, raising the possibility that a legitimate-looking account is later altered to contain misleading information, false brokerage affiliations, or inaccurate credentials.

On the positive side, basic script injection attempts did not successfully execute, and account management functionality appeared organized into structured sections for profile, password, and account resources. It was not possible, however, to determine the full extent of the platform's verification process from external testing alone.

Recommendation

Providing public information about agent verification procedures would further strengthen user confidence in the platform.

Application Security Assessment

Testing focused on publicly accessible forms, user input handling, and frontend validation, including the Contact Form, Agent Registration Form, and Password Recovery workflow.

Browser-side validation was implemented across multiple forms: submissions with missing required fields were blocked, and users were prompted to complete the missing information. A range of malformed email inputs were correctly rejected, demonstrating basic frontend input validation.

The Contact Form accepted legitimate submissions and displayed clear confirmation messaging, with required-field enforcement and email validation holding consistently throughout testing.

A standard XSS payload submitted through the contact form was accepted and displayed a normal confirmation message, but no JavaScript execution occurred — suggesting frontend sanitization or output encoding is already implemented. Because backend administrative systems and internal database rendering were outside the scope of testing, it was not possible to confirm whether user-supplied input is sanitized before storage, administrative review, or email transmission.

Continued sanitization and encoding of user-generated content prior to storage, display rendering, email transmission, administrative review, and database processing is recommended to reduce the likelihood of future injection-based vulnerabilities as the platform expands.

Website Security Configuration Assessment

A browser security header assessment was conducted to evaluate the platform's implementation of standard web security protections. The platform received an overall score of C– (45/100).

Control	Observed Status
HTTPS Enforcement / Redirect	Implemented
HTTP Strict Transport Security (HSTS)	Implemented
Secure / HttpOnly / SameSite Cookie Flags	Implemented
CORS Configuration	Appropriately Restricted
Content Security Policy (CSP)	Not Implemented
X-Frame-Options	Not Implemented
X-Content-Type-Options	Not Implemented
Subresource Integrity (SRI)	Not Implemented

HTTPS enforcement was correctly configured, with traffic redirected to HTTPS and encrypted communications maintained throughout testing. HSTS was enabled, reducing exposure to SSL-stripping attacks. Cookie security demonstrated positive configuration (Secure, HttpOnly, and SameSite flags), and no overly permissive CORS policies were identified.

A Content Security Policy (CSP) was not detected. CSP is one of the most effective browser-level mechanisms for reducing the impact of XSS, unauthorized script execution, malicious third-party script injection, and client-side content manipulation; its absence does not represent an immediate vulnerability but reduces defensive layering against future injection attacks.

X-Frame-Options was also not observed, increasing theoretical susceptibility to clickjacking, and X-Content-Type-Options was missing, which otherwise helps prevent content-sniffing risks. External resources loaded over HTTPS did not appear to use Subresource Integrity (SRI), which would help validate third-party script integrity before execution.

Risk Analysis

The trust-based nature of the platform elevates the business impact of several findings identified during testing. The most significant concern is insufficient realtor identity verification and onboarding governance: because users can create publicly visible realtor accounts without visible verification, the platform faces elevated exposure to impersonation, fraudulent account creation, misleading professional representation, and reputational damage.

Unlike a standard social platform, this platform markets itself as a professional service connecting consumers with trusted real estate professionals, so weaknesses affecting account legitimacy carry outsized business impact. Publicly searchable, unverified profiles increase the likelihood of consumer misinformation, fraudulent representation, fake accounts, spam, brand reputation degradation, and reduced trust in platform recommendations.

ID	Finding	Category	Severity	Status
F-01	No mandatory email verification prior to account activation	Identity & Access	High	Open
F-02	Unverified realtor accounts are publicly searchable immediately after creation	Identity & Access	High	Open
F-03	Licensing and brokerage information editable without validation	Data Integrity	Medium	Open
F-04	No visible rate limiting or lockout on password reset / login	Authentication	Medium	Open
F-05	No multi-factor authentication observed	Authentication	Medium	Open
F-06	Content-Security-Policy header not implemented	Web Configuration	Medium	Open
F-07	X-Frame-Options and X-Content-Type-Options headers missing	Web Configuration	Low	Open
F-08	No Subresource Integrity (SRI) on externally hosted scripts	Web Configuration	Low	Open

ID	Finding	Category	Severity	Status
F-09	No formal, publicly documented vulnerability disclosure process	Governance	Low	Open

No evidence of active exploitation was identified during testing, and no successful cross-site scripting or injection exploitation was observed — a positive indicator, though deeper application testing would be required to fully validate input sanitization across all components. Overall, the current risk profile is best categorized as moderate, with elevated business-trust concerns tied specifically to identity verification and realtor onboarding.

Recommendations

The strongest area for improvement is strengthening realtor identity verification and onboarding controls.

- Require mandatory email verification before account activation or public profile visibility, to reduce fraudulent or disposable account creation
- Withhold public searchability of realtor profiles until professional verification or administrative approval has been completed
- Validate license information against state licensing databases, or review manually before approval; verify brokerage affiliations
- Introduce profile verification indicators or badges so consumers can distinguish verified professionals from newly created accounts
- Require review workflows, logging, or secondary verification for sensitive profile changes (license information, brokerage affiliation, name, service area)
- Implement rate limiting on account creation, login-attempt monitoring, suspicious activity detection, and password-reset abuse protections
- Add administrative audit logging and account review/moderation workflows
- Continue investing in input validation and secure coding practices as functionality expands
- Enforce file-type restrictions, size limitations, malware scanning, secure storage handling, and sanitization controls for uploaded media
- Implement missing browser security headers: Content-Security-Policy, X-Frame-Options, X-Content-Type-Options, and Subresource Integrity for third-party scripts
- Publish a formal vulnerability disclosure process and consider periodic third-party security reviews as the platform scales

Conclusion

The platform demonstrates a functional and organized foundation, with an active social presence, structured account functionality, and partially implemented security protections. Positive indicators observed during testing included organized account management workflows and unsuccessful execution of basic script injection attempts.

The assessment also identified important weaknesses involving realtor identity verification, onboarding governance, and public trust controls. The ability to create publicly searchable realtor profiles without visible verification or approval mechanisms introduces meaningful operational and reputational risk for a platform centered on professional trust and consumer relationships.

At its current stage, the platform is operationally functional but would benefit substantially from stronger identity assurance procedures, onboarding validation workflows, and account governance protections. Implementing stronger verification requirements, approval processes, and monitoring controls would significantly improve both the platform's security posture and overall consumer confidence moving forward.